

Sophie Choi

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EDUCATION

University of California, Irvine

Irvine, CA

Bachelor of Science in Computer Science, GPA: 3.80

WORK EXPERIENCE

Software Developer Intern

Jun 2026 – Aug 2026

Naval Air Warfare Center Weapons Division (NAWCWD)

Point Mugu, CA

- Co-authored a bidirectional real-time **C#/.NET** application (**WPF, MVVM**) that streamlines range testing operations by parsing live telemetry packets into human-readable format and transmitting positioning commands to reposition a \$6M radar system in real time
- Developed a system configuration parser to validate network connections at startup and authored data structures that bridge frontend display and backend logic, reducing integration surface errors across the stack
- Streamlined frontend development by iterating GUI layouts (**XAML**) using **user flows, data flows, and UML class diagrams**, enabling confident presentation of design decisions at critical design reviews with stakeholders

EXPERIENCE

Infrastructure Lead

Oct 2025 – Present

Cyber @ UCI Infrastructure Team

Irvine, CA

- Implemented **SSO (Authentik), virtualization (Proxmox), containerization (Docker), and networking** within a production environment serving 100+ total club members
- Reduced cluster overhead by 25% and improved operational efficiency by decommissioning legacy servers and reorganizing **NFS storage configurations**
- Improved storage throughput to 40 GB/s by configuring a dedicated storage switch connecting **Proxmox** nodes to a **TrueNAS R730xd**, isolating storage traffic from the production network
- Leading and designing system architecture and UI/UX workflows (**Figma**) for 4 internal tooling softwares, such as an internal VM provisioning tool that abstracts complex infrastructure workflows for ~20 internal club members
- Deployed Proxmox VE high-availability clustering and load balancing, improving VM failover reliability by eliminating single points of failure across the cluster

Full-Stack Developer

Oct 2025 – Present

Commit the Change

Irvine, CA

- Programmed a full-stack application using **React, Express.js, PostgreSQL, and CSS**, enabling scalable data management and real-time program oversight for a global NPO operating in 5 continents
- Deployed **RESTful APIs in Express.js** with **full CRUD functionality** and Promise-based concurrent data fetching, streamlining frontend-backend data flow and improving platform performance by ~25%
- Collaborated in Agile sprints with pair programming and client feedback to ship features against 2-week deadlines

Software Developer

Apr 2026 – Present

Blockchain @ UCI Smart Contract Team

Irvine, CA

- Engineered an AI-powered exploit generation engine using **OpenAI APIs and Anvil local chain**, delivering binary proof-of-concept attacks on live contract state rather than theoretical vulnerability reports
- Built a **Python Slither static analysis module** detecting reentrancy, access control, and logic flaws, structuring findings as JSON to feed a downstream **LangGraph agent pipeline**
- Containerized the platform with **Docker** and designed a shared **Postgres/pgvector** database architecture, enabling consistent multi-developer access to run state and grounding data

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C++, C#, HTML, CSS, XML

Frameworks/Libraries: React, Node.js, Express, PostgreSQL, .NET, WPF, MVVM, Tailwind CSS

Tools: Docker, Proxmox, IaC, Linux, Figma, Firebase, Unity, Microsoft Office 365

Certifications: Cisco Certified Network Associate (CCNA), Girls Who Code: Data Science for Good